

Highlights

- ET-PRL reformulates HVAC action triggering as online anomaly detection.
- Multi-scale anomaly scoring with local-global thresholds enables dynamic event gating.
- Predictive RL is coupled with event triggering to reduce redundant control updates.
- Real chiller data show ET-PRL outperforms fixed-step RL, with 53.54% fewer updates and 2.13% lower energy consumption.